

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5 Total Marks: 50

Annexure A

Scenario-Based

Code of Conduct:

I Jayashree M (192524373) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Harinigopi

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25 School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

SOLUTION:

Given

- Network Address: 192.168.100.0/24
- School of Computing → /25
- School of Electronics → /26
- School of Mechanical Engineering → /27

School of Computing (/25)

Subnet Mask = 255.255.255.128

Total Addresses =

Usable Hosts = $128 - 2 = 126$

- Network Address: 192.168.100.0
- Broadcast Address: 192.168.100.127
- Valid Host Range: 192.168.100.1 – 192.168.100.126
- Usable Hosts: 126

School of Electronics (/26)

Subnet Mask = 255.255.255.192

Total Addresses =

Usable Hosts = $64 - 2 = 62$

- Network Address: 192.168.100.128
- Broadcast Address: 192.168.100.191
- Valid Host Range: 192.168.100.129 – 192.168.100.190
- Usable Hosts: 62

School of Mechanical Engineering (/27)

Subnet Mask = 255.255.255.224

Total Addresses =

Usable Hosts = $32 - 2 = 30$

- Network Address: 192.168.100.192
- Broadcast Address: 192.168.100.223
- Valid Host Range: 192.168.100.193 – 192.168.100.222
- Usable Hosts: 30

Remaining Unused IP Address Range

- Network Address: 192.168.100.224/27
- Valid Host Range: 192.168.100.225 – 192.168.100.254
- Broadcast Address: 192.168.100.255
- Usable Hosts: 30

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24 Quality Assurance → /25 Human Resources → /26 Executive Management → /27 As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each department.

2. Identify the valid range of host IP addresses.

3. Calculate the number of usable hosts in each subnet.

4. Identify the unused IP address range remaining within the allocated IP block.

SOLUTION:

Given

- IP Block: 10.10.0.0/16
- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

Software Development (/24)

Subnet Mask = 255.255.255.0

Total Addresses =

Usable Hosts = $256 - 2 = 254$

- Network Address: 10.10.0.0
- Broadcast Address: 10.10.0.255
- Valid Host Range: 10.10.0.1 – 10.10.0.254
- Usable Hosts: 254

Quality Assurance (/25)

Subnet Mask = 255.255.255.128

Total Addresses =

Usable Hosts = $128 - 2 = 126$

- Network Address: 10.10.1.0
- Broadcast Address: 10.10.1.127
- Valid Host Range: 10.10.1.1 – 10.10.1.126
- Usable Hosts: 126

Human Resources (/26)

Subnet Mask = 255.255.255.192

Total Addresses =

Usable Hosts = $64 - 2 = 62$

- Network Address: 10.10.1.128
- Broadcast Address: 10.10.1.191
- Valid Host Range: 10.10.1.129 – 10.10.1.190
- Usable Hosts: 62

Executive Management (/27)

Subnet Mask = 255.255.255.224

Total Addresses =

Usable Hosts = $32 - 2 = 30$

- Network Address: 10.10.1.192
- Broadcast Address: 10.10.1.223
- Valid Host Range: 10.10.1.193 – 10.10.1.222
- Usable Hosts: 30

Remaining Unused IP Address Range

Unused Range: 10.10.1.224 – 10.10.255.255

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network: 2001:0db8:abcd:0000:0000:0000:0000:0000/64. As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**
- 3. Identify the network prefix using the /64 prefix length.**
- 4. Calculate the total number of host addresses supported within the subnet.**
- 5. the total number of possible host addresses available within this network.**

SOLUTION:

Given

IPv6 Address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

Compressed IPv6 Address

Replace consecutive 0000 blocks with ::

Compressed Address: 2001:db8:abcd::/64

Expanded IPv6 Address

Expand the compressed address back to 8 groups of 4 hexadecimal digits.

Expanded Address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

Network Prefix (/64)

A /64 prefix means the first 64 bits represent the network.

Network Prefix: 2001:0db8:abcd:0000::/64

Total Host Addresses Supported

Host bits = $128 - 64 = 64$

Total Host Addresses: 18,446,744,073,709,551,616

Total Possible Host Addresses Available

$2^{64} = 18,446,744,073,709,551,616$ addresses

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries: 192.168.1.0/24 192.168.2.0/24 192.168.3.0/24. A packet arrives at the router with the destination IP address: 192.168.2.45. As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.**
- 2. The matching routing table entry.**
- 3. The next-hop network/interface used to forward the packet.**
- 4. The default route that should be used if the destination does not match any routing table entry.**

SOLUTION:

Given

Routing Table:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP Address: 192.168.2.45

Destination Subnet

The IP address 192.168.2.45 belongs to the subnet:

192.168.2.0/24

Matching Routing Table Entry

Matching Entry: 192.168.2.0/24

Next-Hop Network / Interface

The packet is forwarded through the interface (or next-hop) associated with the 192.168.2.0/24 network.

Default Route

If no routing table entry matches the destination IP, the router forwards the packet using the default route:

0.0.0.0/0

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows: Administration LAN → 192.168.10.0/26 Finance LAN → 192.168.10.64/26 The router interfaces are configured with: Administration Gateway → 192.168.10.1 Finance Gateway → 192.168.10.65 As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.**
- 2. Identify the valid host IP address range.**
- 3. Suggest suitable IP addresses for three computers in each LAN.**
- 4. Identify the default gateway that hosts in each LAN should use for communication.**

SOLUTION:

Given

- Administration LAN → **192.168.10.0/26**
- Finance LAN → **192.168.10.64/26**
- Administration Gateway → **192.168.10.1**
- Finance Gateway → **192.168.10.65**

Administration LAN (/26)

Subnet Mask = **255.255.255.192**

Total Addresses =

Usable Hosts = **64 – 2 = 62**

- **Network Address:** 192.168.10.0
- **Broadcast Address:** 192.168.10.63
- **Valid Host Range:** 192.168.10.1 – 192.168.10.62
- **Three Computer IPs:** 192.168.10.2, 192.168.10.3, 192.168.10.4
- **Default Gateway:** 192.168.10.1

Finance LAN (/26)

Subnet Mask = **255.255.255.192**

Total Addresses =

Usable Hosts = **64 – 2 = 62**

- **Network Address:** 192.168.10.64
- **Broadcast Address:** 192.168.10.127
- **Valid Host Range:** 192.168.10.65 – 192.168.10.126
- **Three Computer IPs:** 192.168.10.66, 192.168.10.67, 192.168.10.68
- **Default Gateway:** 192.168.10.65

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand